HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

Rum

OSA Digital Twin

Predicting obstructive sleep apnea events by fusing a synthetic EHR risk profile with wearable-style physiological signals.

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01 The Problem: OSA Is Common and Badly Underdiagnosed

Obstructive Sleep Apnea — a chronic, lifestyle-linked condition

Why OSA

- Closely tied to obesity, hypertension and type 2 diabetes — all rising in India.
- Diagnosis normally needs an overnight sleep-lab study, which most patients never get.
- A Digital Twin can estimate risk continuously, without waiting for a formal diagnosis.
- This project builds a proof-of-concept: fuse a patient's health history with real physiological signal to flag risk, window by window, through the night.

What "Digital Twin" Means Here

- Static data: a synthetic EHR — comorbidities such as obesity, hypertension, diabetes.
- Dynamic data: overnight ECG and respiration — wearable-style, from real anonymized recordings.
- Fused output: a single risk flag (LOW / MODERATE / HIGH) a clinician could act on.

2

data streams fused into one twin

3

risk tiers — LOW / MODERATE / HIGH

0

real patient records required

Rum — OSA Digital Twin: System Architecture

A synthetic EHR risk profile and real wearable-style physiological signals are fused into one clinical flag.

STATIC (patient history)

Synthea

Open-source synthetic patient generator — fully synthetic EHR, zero real patient data.

↓

Weighted comorbidity score

Rule-weighted: obesity, hypertension, prediabetes, type 2 diabetes, existing sleep apnea diagnosis.

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Static risk (cohort percentile, 0–1)

Patient's rank within the synthetic population's weighted risk score.

DYNAMIC (real-time signal)

MIT-BIH Polysomnographic Database

PhysioNet, open-access. Real ECG + nasal airflow + SpO2, expert-scored apnea/hypopnea annotations.

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30-second windows → HRV + breathing features

Heart rate, RR-interval SDNN, LF/HF ratio, breath rate, breath-interval variability, breath amplitude.

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XGBoost + causal smoothing

Leave-one-patient-out validated. Trailing 3-min average → dynamic risk score (0–1).

↘ ↙

Fusion

risk = 0.3 × static risk + 0.7 × dynamic risk (hand-set weights, illustrative)

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Threshold flag

LOW (<0.50) **MODERATE** (0.50-0.65) **HIGH** (≥0.65)

↓

Clinician Dashboard (concept)

Patients ranked by fused risk — overnight risk timeline — EHR risk tags. Not a diagnostic device.

03 Data Sources

Open-access only — no request, approval, or credentialing required

MIT-BIH Polysomnographic Database

- PhysioNet, Open Data Commons Attribution License.
- Real ECG, nasal airflow and SpO2 signals.
- Expert-scored sleep-stage AND apnea/hypopnea annotations.
- 15 of 18 subjects used — selected for a consistent nasal-airflow channel and a spread of severities, plus 2 zero-event healthy controls.

Synthea

- Open-source synthetic patient generator.
- Fully synthetic EHR — zero real patient data.
- Filtered for OSA-relevant comorbidities: obesity, hypertension, prediabetes, type 2 diabetes.
- Explicitly named as an acceptable EHR source in the challenge brief.

18

subjects available, fully open-access

15

used in this prototype

300

synthetic patients generated

04 Methodology Highlights

Decisions made along the way, and why

1 30-second windows, real labels

Each overnight recording is cut into 30s windows aligned to expert annotations — not self-invented proxy labels.

2 Caught a data-leakage mistake

A random train/test split scored 0.94 AUC. Splitting by patient (Leave-One-Patient-Out) revealed the true score: 0.65.

3 Reframed as early warning

Beyond “is this window an event,” the model predicts whether an event occurs in the next ~90 seconds.

4 Checked against a shortcut

A persistence baseline (just repeat the current label) scored 0.776 — higher than the naive predictive model. The honest metric only counts currently-normal windows.

5 Smoothing + threshold calibration

A causal 3-minute trailing average, with thresholds chosen from out-of-fold predictions and checked against 2 zero-event healthy controls for false alarms.

05 Results

Leave-one-patient-out ROC-AUC — 13 scoreable subjects

Task	Mean ROC-AUC
Detect event in the current window	0.687
Predict event within ~90s, all windows	0.722
Persistence baseline (current label only)	0.776
Predict event within ~90s, currently-normal windows only (headline result)	**0.658**
Onset-only, 4 SpO2 subjects — without SpO2	0.660
Onset-only, 4 SpO2 subjects — with SpO2	0.730

Why 0.658 is the headline number: apnea events cluster, so "current label predicts future label" alone scores 0.776 — higher than the naive predictive model (0.722). The 0.658 figure removes that shortcut by testing only windows that are currently normal.

06 Alert Thresholds & Fusion in Action

Same physiological signal, different patients, different urgency

Threshold	Alert rate	Precision	Recall	False-alarm (healthy)
HIGH ≥ 0.65	22%	0.86	0.41	10%
MODERATE ≥ 0.50	39%	0.70	0.59	37%

Thresholds chosen from out-of-fold predictions, validated against 2 zero-event healthy controls as a false-alarm check.

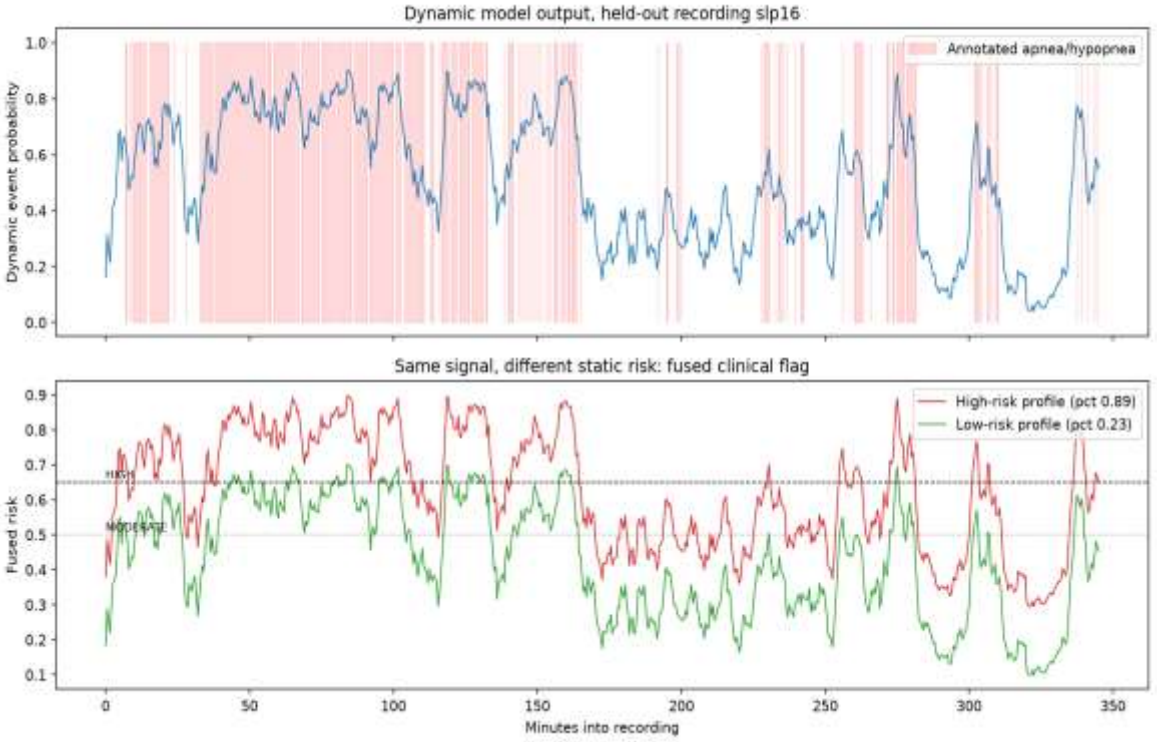
46%

HIGH flag rate — patient at the 89th EHR-risk percentile

8%

HIGH flag rate — patient at the 23rd EHR-risk percentile

Same recording (slp16, 51 events/hr). Only the static EHR profile changed.



Held-out recording slp16: dynamic model output (top) and the fused flag for a high-risk vs low-risk EHR profile (bottom).

07 Limitations & Future Work

Known gaps, stated plainly

- Modest absolute performance (onset-only AUC ~0.66). 30-second windows are short for frequency-domain HRV, which conventionally uses 2–5 minutes.
- Hyperparameters were tuned on all subjects before leave-one-patient-out evaluation, so scores carry mild optimism. Nested cross-validation is the fix.
- The SpO2 experiment covers only 4 severe-apnea subjects with no controls; “normal” windows may include recovery from a preceding event.
- Real signals and synthetic EHR cannot be linked per patient (privacy constraint) — fusion happens at the score level, with hand-set, illustrative weights (0.3 / 0.7) and thresholds, not clinically calibrated.
- Future work: longer windows, sequence models on raw signals, probability calibration, a larger and more diverse cohort, and prospective validation on consumer wearables.
- This is a research prototype, not a diagnostic device.

Thank You

Team Rum — OSA Digital Twin

GitHub: https://github.com/swe74-29/Rum_IGDTUW

Research prototype — not a diagnostic device.